Section 3.1: Partial and Directional Derivatives

Based on *A Course in Calculus and Real Analysis*
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Outline

Motivation

- In multivariable calculus, functions often depend on several variables:

$$f : \mathbb{R}^n \to \mathbb{R}, \quad f(x_1, x_2, \ldots, x_n).$$

- We study how f changes when one variable varies while others are fixed — leading to the notion of **partial derivatives**.
- Later, we consider the rate of change of f along an arbitrary direction — the **directional derivative**.

Definition of Partial Derivative

Let $f(x, y)$ be a real function of two variables defined on a domain $D \subseteq \mathbb{R}^2$.

Partial Derivative with respect to x at (a, b)

$$\frac{\partial f}{\partial x}(a, b) = \lim_{h \to 0} \frac{f(a+h, b) - f(a, b)}{h},$$

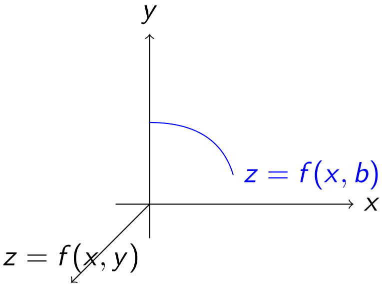
if the limit exists.

Similarly,

$$\frac{\partial f}{\partial y}(a, b) = \lim_{h \to 0} \frac{f(a, b+h) - f(a, b)}{h}.$$

Interpretation

- $\dfrac{\partial f}{\partial x}(a, b)$ measures the instantaneous rate of change of f when x changes at (a, b) while y is held fixed.
- Similarly for $\dfrac{\partial f}{\partial y}(a, b)$.
- Geometrically, these correspond to slopes of tangent lines to the surface $z = f(x, y)$ in coordinate directions.



Example

Let $f(x, y) = x^2y + 3xy^2$.

$$f_x = 2xy + 3y^2, \quad f_y = x^2 + 6xy.$$

At $(1, 1)$:

$$f_x(1, 1) = 5, \quad f_y(1, 1) = 7.$$

Hence, f increases faster in the y-direction than in the x-direction near $(1, 1)$.

Mean Value Theorem for Functions of Several Variables

Theorem (Directional Form)

Let $f : D \subseteq \mathbb{R}^n \to \mathbb{R}$ be differentiable on an open convex set D. For any $\mathbf{a}, \mathbf{b} \in D$, there exists $\mathbf{c}$ on the line segment joining $\mathbf{a}$ and $\mathbf{b}$ such that

$$f(\mathbf{b}) - f(\mathbf{a}) = \nabla f(\mathbf{c}) \cdot (\mathbf{b} - \mathbf{a}).$$

Interpretation:

- Generalizes the single-variable result

$$f(b) - f(a) = f'(c)(b - a)$$

  to higher dimensions.
- The derivative $f'(c)$ becomes the gradient $\nabla f(\mathbf{c})$.

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Proof Idea

- Define a single-variable function:

$$\phi(t) = f(\mathbf{a} + t(\mathbf{b} - \mathbf{a})), \quad t \in [0, 1].$$

- Then ϕ is differentiable and by 1D MVT,

$$\phi(1) - \phi(0) = \phi'(t_0)$$

  for some $t_0 \in (0, 1)$.

- But

$$\phi'(t) = \nabla f(\mathbf{a} + t(\mathbf{b} - \mathbf{a})) \cdot (\mathbf{b} - \mathbf{a}).$$

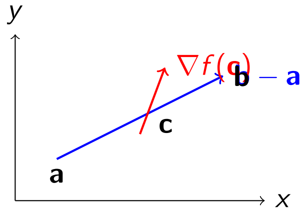
  Hence,

$$f(\mathbf{b}) - f(\mathbf{a}) = \nabla f(\mathbf{c}) \cdot (\mathbf{b} - \mathbf{a}),$$

  where $\mathbf{c} = \mathbf{a} + t_0(\mathbf{b} - \mathbf{a})$.

Geometric Meaning

- The difference $f(\mathbf{b}) - f(\mathbf{a})$ equals the projection of $\nabla f(\mathbf{c})$ along $\mathbf{b} - \mathbf{a}$.
- In $\mathbb{R}^2$, this means there exists a point $\mathbf{c}$ between $\mathbf{a}$ and $\mathbf{b}$ where the tangent plane's slope in that direction equals the average rate of change.



Example

Let $f(x,y) = x^2y + y^3$ and $\mathbf{a} = (0,0)$, $\mathbf{b} = (1,1)$.

$$f(\mathbf{b}) - f(\mathbf{a}) = (1^2)(1) + 1^3 - 0 = 2.$$

We need $\mathbf{c}$ such that

$$f(\mathbf{b}) - f(\mathbf{a}) = \nabla f(\mathbf{c}) \cdot (\mathbf{b} - \mathbf{a}).$$

$$\nabla f(x,y) = (2xy, x^2 + 3y^2).$$

On line $y = x$, $\nabla f(x,x) = (2x^2, x^2 + 3x^2) = (2x^2, 4x^2)$.

$$\nabla f(x,x) \cdot (1,1) = 6x^2.$$

Thus $2 = 6c^2 \Rightarrow c = \sqrt{\frac{1}{3}}$.

$$\mathbf{c} = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right).$$

Concept of Directional Derivative

Partial derivatives measure rate of change along coordinate axes only. For an arbitrary direction $\mathbf{u} = (u_1, u_2)$, with $\|\mathbf{u}\| = 1$, we define the **directional derivative** of f at (a, b) along $\mathbf{u}$ by:

$$D_{\mathbf{u}} f(a, b) = \lim_{h \to 0} \frac{f((a, b) + h\mathbf{u}) - f(a, b)}{h},$$

if the limit exists.

Relation to Partial Derivatives

If f has partial derivatives at (a, b), then

$$D_{\mathbf{u}} f(a, b) = f_x(a, b)u_1 + f_y(a, b)u_2.$$

Gradient Vector

$$\nabla f(a, b) = (f_x(a, b), f_y(a, b)).$$

Then

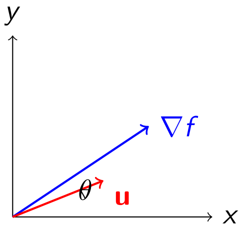
$$D_{\mathbf{u}} f(a, b) = \nabla f(a, b) \cdot \mathbf{u}.$$

Geometric Meaning

- The gradient vector $\nabla f(a, b)$ points in the direction of the maximum rate of increase of f.
- The rate of change of f in the direction $\mathbf{u}$ is given by the projection of ∇f onto $\mathbf{u}$:

$$D_{\mathbf{u}} f(a, b) = |\nabla f(a, b)| \cos\theta,$$

where θ is the angle between $\nabla f(a, b)$ and $\mathbf{u}$.



Example

Let $f(x,y) = x^2y + y^3$.

$$\nabla f = (2xy, x^2 + 3y^2).$$

At $(1,2)$, $\nabla f(1,2) = (4,13)$.
For direction $\mathbf{u} = \frac{1}{\sqrt{2}}(1,1)$:

$$D_{\mathbf{u}} f(1,2) = \frac{1}{\sqrt{2}}(4+13) = \frac{17}{\sqrt{2}}.$$

Thus, f increases most rapidly in the direction of $\nabla f = (4,13)$.

Generalization to n Variables

If $f : \mathbb{R}^n \to \mathbb{R}$ is differentiable at $\mathbf{a}$, then

$$D_{\mathbf{u}} f(\mathbf{a}) = \nabla f(\mathbf{a}) \cdot \mathbf{u},$$

where $\nabla f(\mathbf{a}) = \left(\frac{\partial f}{\partial x_1}(\mathbf{a}), \ldots, \frac{\partial f}{\partial x_n}(\mathbf{a}) \right)$ and $\mathbf{u}$ is a unit vector in $\mathbb{R}^n$.
The maximum directional derivative equals $|\nabla f(\mathbf{a})|$.

Proposition 3.9

Statement

Let $f : D \subseteq \mathbb{R}^2 \to \mathbb{R}$ have partial derivatives at (a, b). Then for any unit vector $\mathbf{u} = (u_1, u_2)$, the directional derivative of f at (a, b) along $\mathbf{u}$ exists and

$$D_{\mathbf{u}} f(a, b) = f_x(a, b)u_1 + f_y(a, b)u_2.$$

Remarks:

- The existence of partial derivatives implies existence of directional derivatives in all directions.
- The formula is linear in $\mathbf{u}$.

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Example (Proposition 3.9)

Let $f(x, y) = x^2 y + y^3$.

$$f_x = 2xy, \quad f_y = x^2 + 3y^2.$$

At $(1, 2)$ and direction $\mathbf{u} = \frac{1}{\sqrt{2}}(1, 1)$,

$$D_{\mathbf{u}} f(1, 2) = (2 \cdot 1 \cdot 2)\frac{1}{\sqrt{2}} + (1^2 + 3 \cdot 2^2)\frac{1}{\sqrt{2}} = \frac{4 + 13}{\sqrt{2}} = \frac{17}{\sqrt{2}}.$$

Proposition 3.10

Statement

If f has partial derivatives at (a, b), then

$$D_{\mathbf{u}} f(a, b) = \nabla f(a, b) \cdot \mathbf{u},$$

and hence

$$|D_{\mathbf{u}} f(a, b)| \leq |\nabla f(a, b)|.$$

Equality occurs when $\mathbf{u}$ is parallel to $\nabla f(a, b)$.

Interpretation:

- $\nabla f(a, b)$ gives the direction of steepest ascent.
- The maximum rate of change equals $|\nabla f(a, b)|$.

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Example (Proposition 3.10)

Let $f(x, y) = 3x^2 + 4y^2$. Then $\nabla f(x, y) = (6x, 8y)$.
At $(1, 1)$: $\nabla f(1, 1) = (6, 8)$ and $|\nabla f| = 10$.
Thus, maximum rate of increase $= 10$, in direction $\mathbf{u} = \frac{1}{10}(6, 8)$.

Proposition 3.11

Statement

Let $f, g : D \to \mathbb{R}$ have partial derivatives at (a, b) and $\alpha, \beta \in \mathbb{R}$. Then

$$D_{\mathbf{u}}(\alpha f + \beta g)(a, b) = \alpha D_{\mathbf{u}} f(a, b) + \beta D_{\mathbf{u}} g(a, b).$$

Proof Sketch:

$$\nabla(\alpha f + \beta g) = \alpha \nabla f + \beta \nabla g \Rightarrow D_{\mathbf{u}}(\alpha f + \beta g) = \mathbf{u} \cdot \nabla(\alpha f + \beta g).$$

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Example (Proposition 3.11)

Let $f(x,y) = x^2y$, $g(x,y) = xy^2$, $\mathbf{u} = (1,0)$.

$$D_{\mathbf{u}}f = f_x = 2xy, \quad D_{\mathbf{u}}g = g_x = y^2.$$

Then for $h = 3f + 2g$:

$$D_{\mathbf{u}}h = 3(2xy) + 2(y^2) = 6xy + 2y^2.$$

Proposition 3.12

Statement

If $f, g : D \to \mathbb{R}$ have partial derivatives at (a, b), then:

$$D_{\mathbf{u}}(fg)(a,b) = f(a,b)D_{\mathbf{u}}g(a,b) + g(a,b)D_{\mathbf{u}}f(a,b).$$

If $g(a,b) \neq 0$, then

$$D_{\mathbf{u}}\left(\frac{f}{g}\right)(a,b) = \frac{g(a,b)D_{\mathbf{u}}f(a,b) - f(a,b)D_{\mathbf{u}}g(a,b)}{[g(a,b)]^2}.$$

Example (Proposition 3.12)

Let $f(x,y) = x$, $g(x,y) = y^2$, $\mathbf{u} = \frac{1}{\sqrt{2}}(1,1)$.

$$f_x = 1, \quad f_y = 0, \quad g_x = 0, \quad g_y = 2y.$$

At $(1,2)$:

$$D_{\mathbf{u}}(fg) = fD_{\mathbf{u}}g + gD_{\mathbf{u}}f = 1\cdot(0\cdot u_1 + 2yu_2) + 4\cdot(1\cdot u_1 + 0) = 4(\tfrac{1}{\sqrt{2}}) + 4(\tfrac{1}{\sqrt{2}}) = \tfrac{8}{\sqrt{2}}.$$

Proposition 3.13

Statement

Let $f : \mathbb{R}^2 \to \mathbb{R}$ have partial derivatives at (a, b) and let $x = x(t), y = y(t)$ be differentiable at t_0. Then

$$\frac{d}{dt} f(x(t), y(t))\Big|_{t=t_0} = f_x(a, b)\frac{dx}{dt}(t_0) + f_y(a, b)\frac{dy}{dt}(t_0),$$

where $(a, b) = (x(t_0), y(t_0))$.

Vector form:

$$\frac{df}{dt} = \nabla f \cdot \frac{d\mathbf{r}}{dt}, \quad \text{where } \mathbf{r}(t) = (x(t), y(t)).$$

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Example (Proposition 3.13)

Let $f(x, y) = x^2 y$, with $x = t^2$, $y = \sin t$.
Then $f(x(t), y(t)) = t^4 \sin t$.

$$\frac{df}{dt} = 4t^3 \sin t + t^4 \cos t.$$

By chain rule:

$$f_x = 2xy, \quad f_y = x^2.$$

At $(x, y) = (t^2, \sin t)$:

$$f_x \frac{dx}{dt} + f_y \frac{dy}{dt} = (2t^2 \sin t)(2t) + (t^4)(\cos t) = 4t^3 \sin t + t^4 \cos t.$$

Same result.

Proposition 3.14 (Mean Value Theorem)

Statement

Let $D \subseteq \mathbb{R}^n$ be an open convex set, and $f : D \to \mathbb{R}$ differentiable. Then for any $\mathbf{a}, \mathbf{b} \in D$, there exists $\mathbf{c}$ on the line segment joining $\mathbf{a}$ and $\mathbf{b}$ such that

$$f(\mathbf{b}) - f(\mathbf{a}) = \nabla f(\mathbf{c}) \cdot (\mathbf{b} - \mathbf{a}).$$

Interpretation: This generalizes the one-dimensional MVT:

$$f(b) - f(a) = f'(c)(b - a).$$

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Example (Proposition 3.14)

Let $f(x, y) = x^2y + y^3$, $\mathbf{a} = (0, 0)$, $\mathbf{b} = (1, 1)$.

$$f(\mathbf{b}) - f(\mathbf{a}) = 2.$$

$$\nabla f(x, y) = (2xy, x^2 + 3y^2).$$

On line $y = x$, $\nabla f(x, x) = (2x^2, 4x^2)$,

$$\nabla f(x, x) \cdot (1, 1) = 6x^2.$$

So $2 = 6c^2 \Rightarrow c = \frac{1}{\sqrt{3}}$.

$$\mathbf{c} = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right).$$

Motivation

Goal: Extend the idea of directional derivative to higher order.
For a differentiable function $f : \mathbb{R}^2 \to \mathbb{R}$, the first directional derivative gives the rate of change of f along direction $\mathbf{u}$.

$$D_{\mathbf{u}} f(\mathbf{a}) = \lim_{h \to 0} \frac{f(\mathbf{a} + h\mathbf{u}) - f(\mathbf{a})}{h}.$$

We now ask: how does *this rate itself* change along the same or another direction?

Definition: Second and Higher Order Directional Derivatives

Definition

Let $f : D \subseteq \mathbb{R}^n \to \mathbb{R}$ and let $\mathbf{u}, \mathbf{v}$ be directions.
If the first directional derivative $D_{\mathbf{v}} f$ exists near $\mathbf{a}$ and is differentiable at $\mathbf{a}$ in the direction $\mathbf{u}$, then the **second order directional derivative** of f at $\mathbf{a}$ in the directions $\mathbf{u}, \mathbf{v}$ is

$$D_{\mathbf{u}} D_{\mathbf{v}} f(\mathbf{a}) = \lim_{h \to 0} \frac{D_{\mathbf{v}} f(\mathbf{a} + h\mathbf{u}) - D_{\mathbf{v}} f(\mathbf{a})}{h}.$$

Higher-order derivatives are defined recursively:

$$D_{\mathbf{u}_1} D_{\mathbf{u}_2} \cdots D_{\mathbf{u}_k} f(\mathbf{a}).$$

If f has continuous second partial derivatives, then mixed directional derivatives are symmetric:

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Higher-order derivatives are defined recursively:

$$D_{\mathbf{u}_1} D_{\mathbf{u}_2} \cdots D_{\mathbf{u}_k} f(\mathbf{a}).$$

If f has continuous second partial derivatives, then mixed directional derivatives are symmetric:

Second Directional Derivative in Terms of Hessian

If f is twice differentiable at $\mathbf{a}$ with Hessian matrix

$$H_f(\mathbf{a}) = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix},$$

then for any direction $\mathbf{u} = (u_1, u_2)$,

$$D_{\mathbf{u}}^2 f(\mathbf{a}) = \mathbf{u}^T H_f(\mathbf{a})\mathbf{u} = f_{xx} u_1^2 + 2 f_{xy} u_1 u_2 + f_{yy} u_2^2.$$

Interpretation:

- Measures the curvature of f along $\mathbf{u}$.
- Positive value f bends upward (convex in that direction).
- Negative value f bends downward.

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- Negative value f bends downward.

Example 1: Second Directional Derivative

Let $f(x,y) = x^2y + y^3$.
Then

$$f_x = 2xy, \quad f_y = x^2 + 3y^2,$$

$$f_{xx} = 2y, \quad f_{xy} = 2x, \quad f_{yy} = 6y.$$

At point $(1,1)$ and direction $\mathbf{u} = \frac{1}{\sqrt{2}}(1,1)$:

$$H_f(1,1) = \begin{bmatrix} 2 & 2 \\ 2 & 6 \end{bmatrix}.$$

$$D_{\mathbf{u}}^2 f(1,1) = \mathbf{u}^T H_f(1,1)\mathbf{u} = \tfrac{1}{2}(2+4+6) = 6.$$

Thus, the second directional derivative along $\mathbf{u}$ is 6.

Example 2: Mixed Directional Derivatives

Let $f(x, y) = x^3y^2$. Compute $D_{\mathbf{i}}D_{\mathbf{j}}f(1, 1)$ and $D_{\mathbf{j}}D_{\mathbf{i}}f(1, 1)$, where $\mathbf{i} = (1, 0)$ and $\mathbf{j} = (0, 1)$.

$$f_x = 3x^2y^2, \quad f_y = 2x^3y.$$

Then

$$D_{\mathbf{i}}D_{\mathbf{j}}f = f_{xy} = 6xy, \quad D_{\mathbf{j}}D_{\mathbf{i}}f = f_{yx} = 6xy.$$

At $(1, 1)$: both equal 6.
Hence $D_{\mathbf{i}}D_{\mathbf{j}}f = D_{\mathbf{j}}D_{\mathbf{i}}f$ when second partials are continuous.

Geometric Interpretation

- The **first directional derivative** gives slope along a direction.
- The **second directional derivative** gives concavity or curvature in that direction.
- Along **u**, $D^2_{\mathbf{u}} f > 0$ bowl-shaped (convex); < 0 cap-shaped (concave).
- The Hessian combines all such curvatures; it defines a quadratic form.

example-graph.png

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- The Hessian combines all such curvatures; it defines a quadratic form.

example-graph.png

Summary of Higher-Order Directional Derivatives

- Defined recursively: $D_{\mathbf{u}_1} D_{\mathbf{u}_2} \dots D_{\mathbf{u}_k} f$.
- For twice-differentiable f, $D^2_{\mathbf{u}} f = \mathbf{u}^T H_f \mathbf{u}$.
- Continuous second partials mixed directional derivatives commute.
- Second directional derivative captures curvature along $\mathbf{u}$.

$$\boxed{D^2_{\mathbf{u}} f = f_{xx} u_1^2 + 2 f_{xy} u_1 u_2 + f_{yy} u_2^2.}$$

Summary of Higher-Order Directional Derivatives

- Defined recursively: $D_{\mathbf{u}_1} D_{\mathbf{u}_2} \dots D_{\mathbf{u}_k} f$.
- For twice-differentiable f, $D_{\mathbf{u}}^2 f = \mathbf{u}^T H_f \mathbf{u}$.
- Continuous second partials mixed directional derivatives commute.
- Second directional derivative captures curvature along $\mathbf{u}$.

$$\boxed{D_{\mathbf{u}}^2 f = f_{xx} u_1^2 + 2 f_{xy} u_1 u_2 + f_{yy} u_2^2.}$$

Summary

- Partial derivatives: rate of change along axes.
- Directional derivative: rate of change along any direction.
- Gradient: vector of partial derivatives.
- ∇f gives:
  - Direction of steepest ascent.
  - Magnitude $=$ maximum rate of change.

References

S.R. Ghorpade and B.V. Limaye, *A Course in Calculus and Real Analysis*, Springer, 2006.